Universidade Leiria • Oeste
Escola de Tecnologia e Gestão

Information Systems Integration

Part 1 – Background

A concrete/real scenario

Background

- Over the years…
  - Companies have been acquiring information technology
    - Technology silos were created inside companies
- Nowadays
  - Competitiveness…
  - Constant market changing…



Companies must be in constant adaptation

Background

Technology silos (today) are the major obstacle to

- Flexibility
- Adaptability



Information generated and maintained in those silos is not shared to meet company new challenges

Systems and applications integration appears to solve this problem

Background

What is Information systems integration?

- Combination of software, hardware and standards to (seamlessly) integrate two or more enterprise systems allowing them to operate as one
- A unified view of the enterprise processes

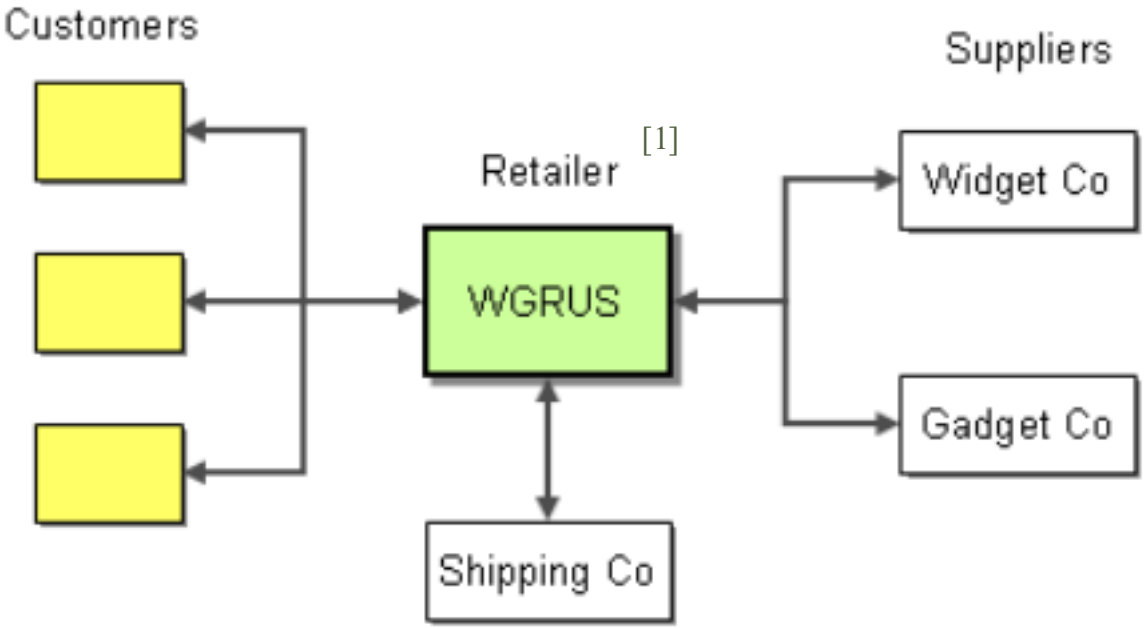


Commonly presented as a web portal, dashboard, etc.

Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

An online retailer that buys widgets and gadgets from manufacturers and resells them to customers



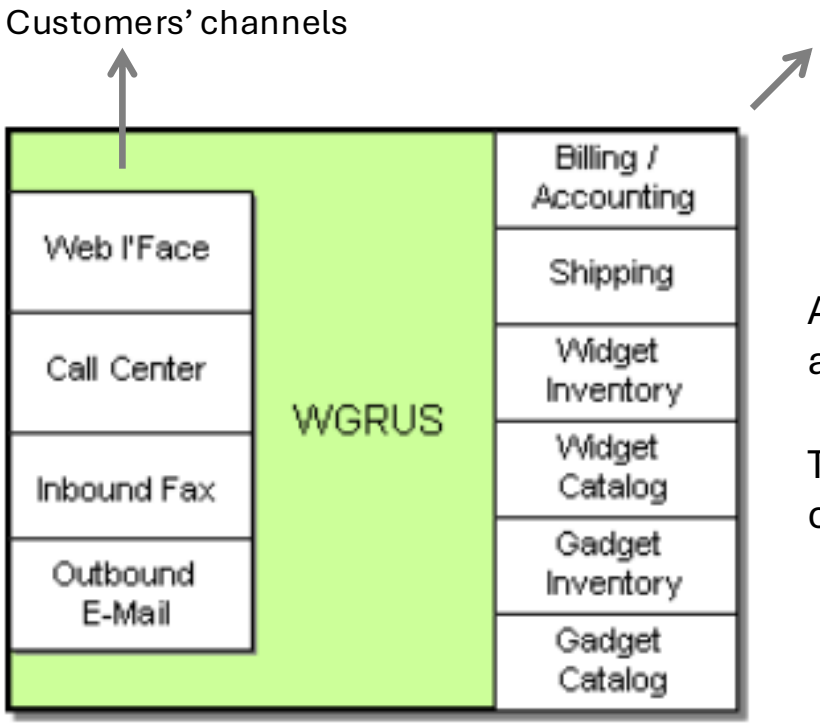
The WGRUS Ecosystem

Kick-off Scenario

- WGRUS - Widgets & Gadgets ‘R Us [1]
  - Expected requirements
    - Take Orders
    - Process Orders
    - Check Status
    - Change Address
    - New Catalog
    - Announcements
    - Testing and Monitoring

Kick-off Scenario

- WGRUS - Widgets & Gadgets ‘R Us [1]
  - Existing internal systems… **(Oops!)**



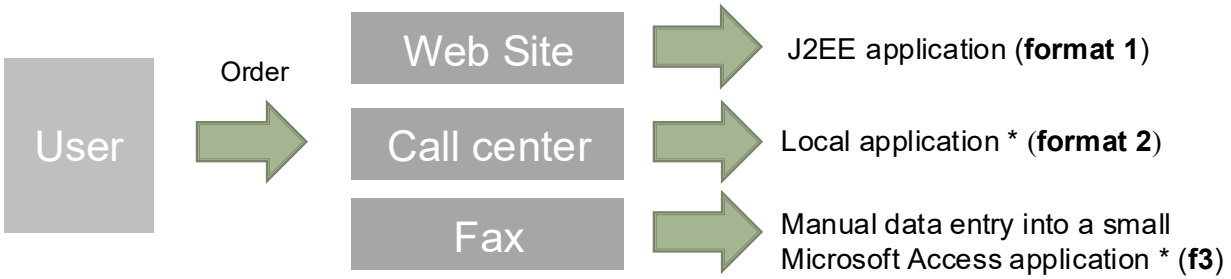
A variety of packaged and custom applications.

This makes integration work challenging

Kick-off Scenario

WGRUS - Widgets & Gadgets 'R Us [1]

- **1. Function Taking Orders**
  - Currently:
    - Tedious manual process
    - Orders below $20 => money loss!



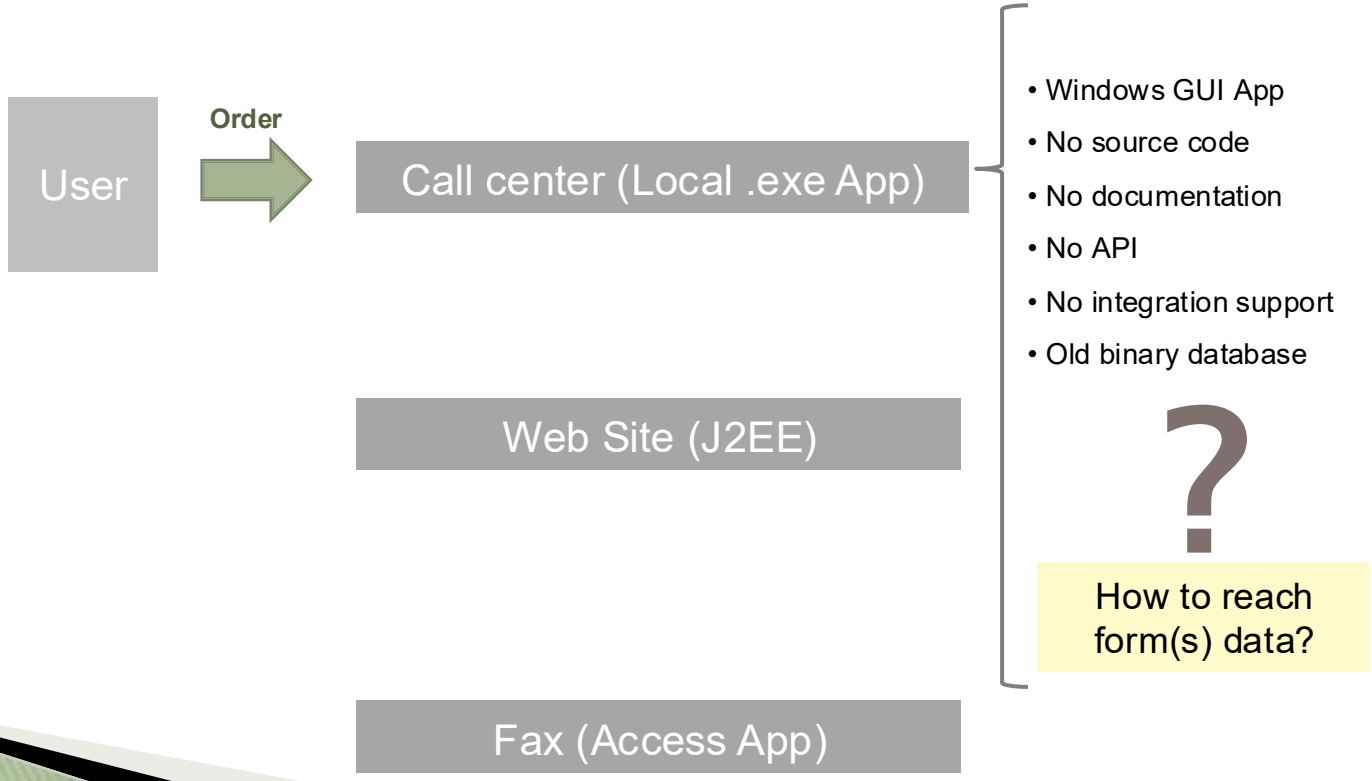
It must be possible to place an order via fax or call center and check its status via web site!

* Developed without integration in mind

Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

1. Function Taking Orders (integrating…)



Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

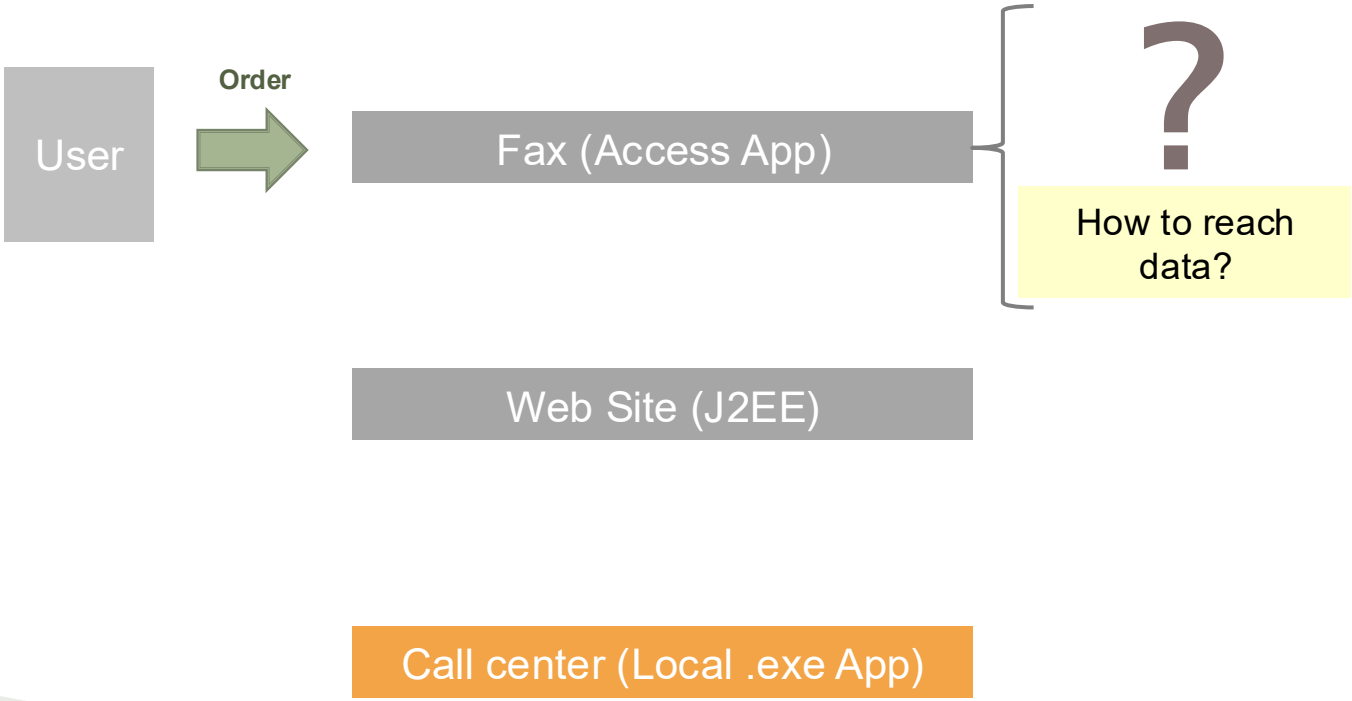
1. Function Taking Orders (integrating…)



Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

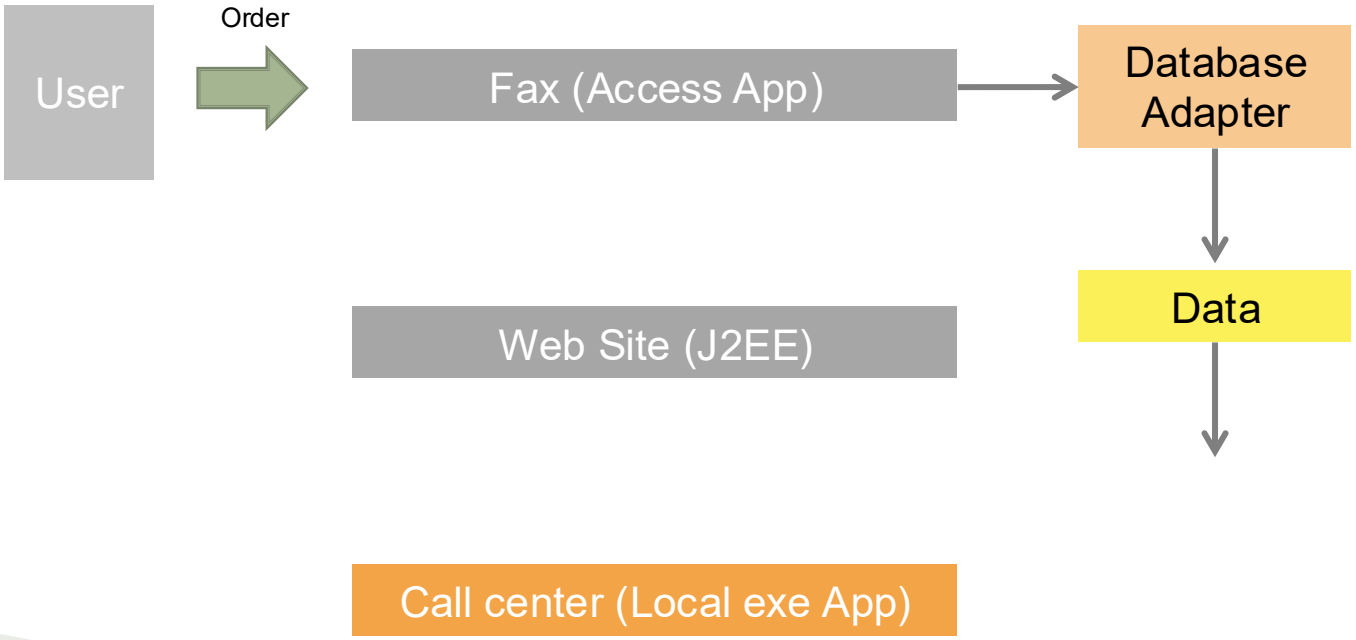
1. Function Taking Orders (integrating…)



Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

1. Function Taking Orders (integrating…)



Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

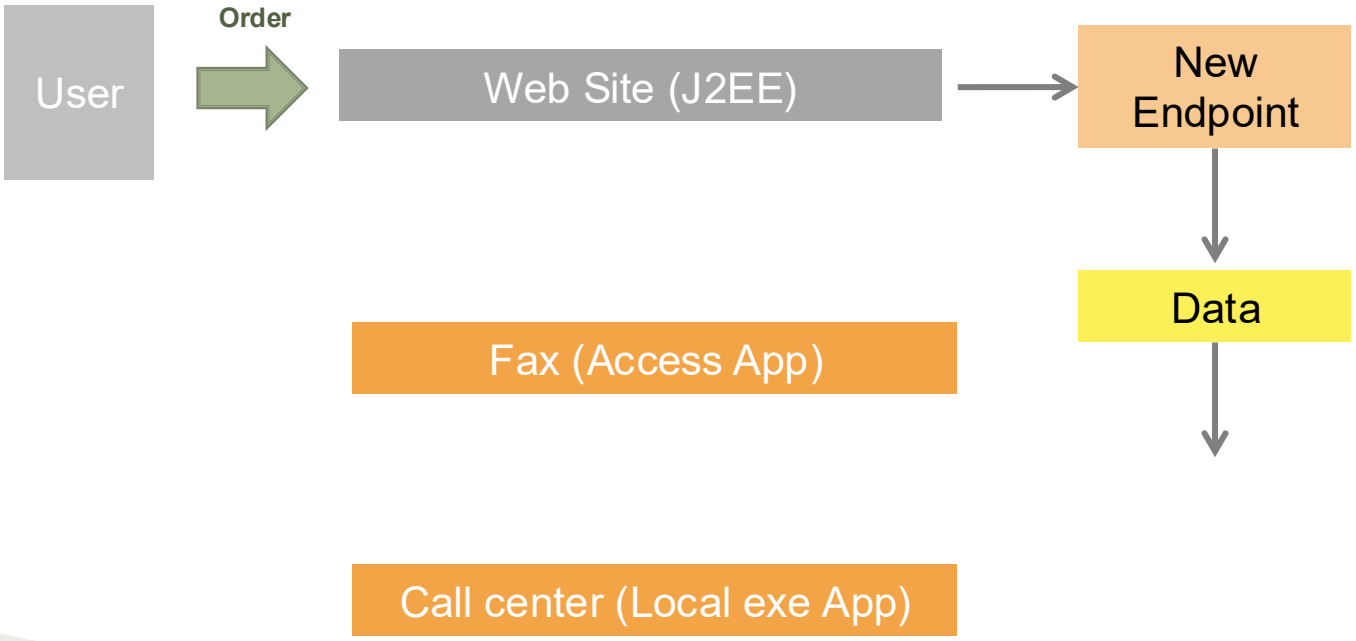
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Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

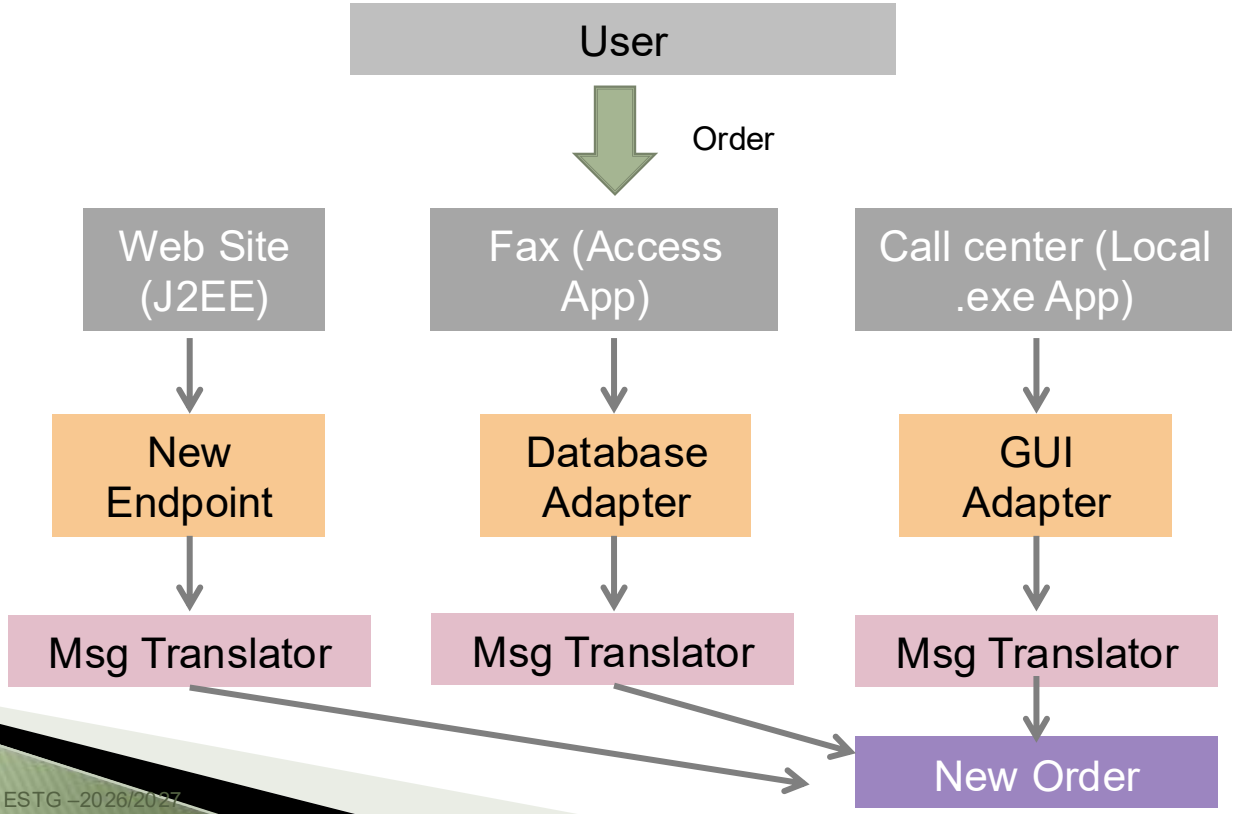
1. Function Taking Orders (integrating…)



Kick-off Scenario

WGRUS - Widgets & Gadgets 'R Us [1]

1. Function Taking Orders (integrating…)



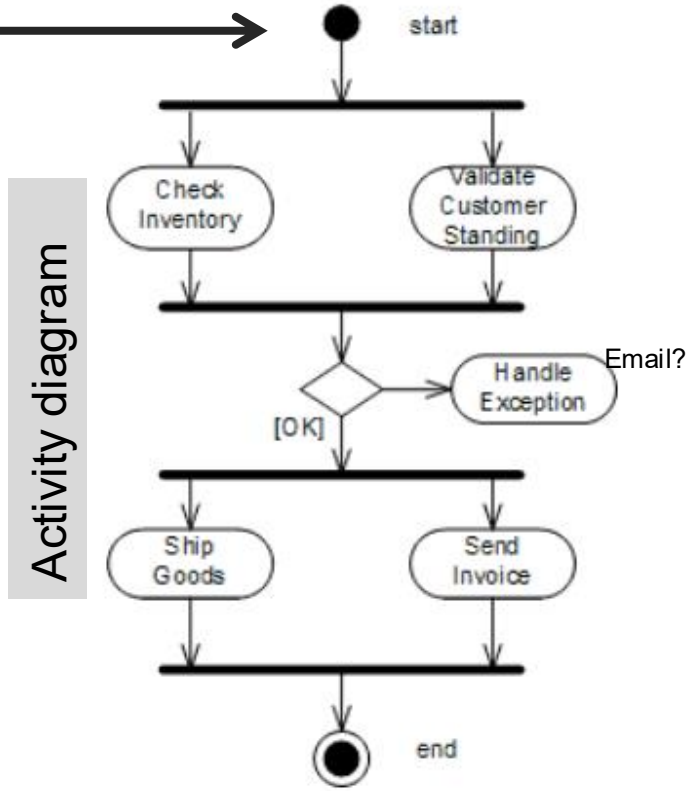
Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

2. Processing Orders

New Order

(Independent of the message source)

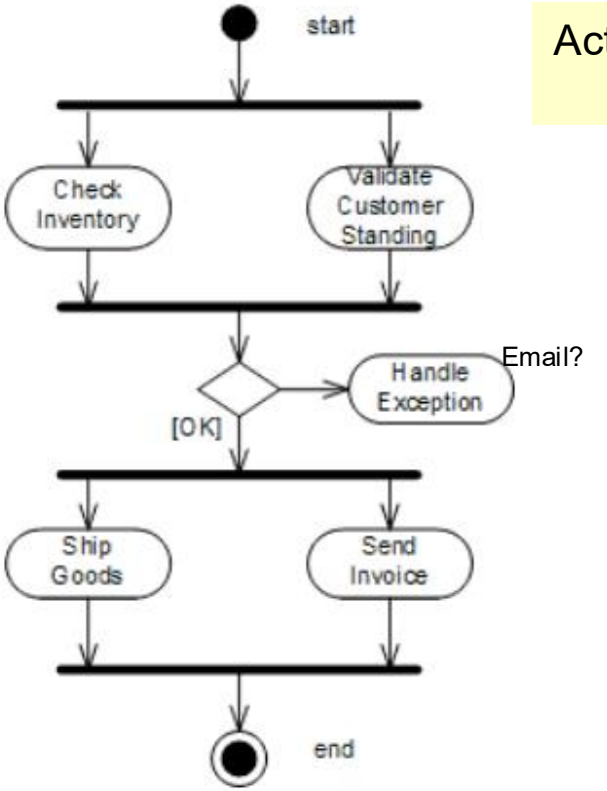


Kick-off Scenario

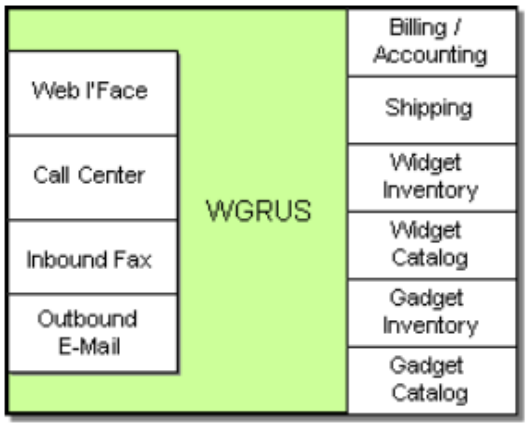
WGRUS - Widgets & Gadgets 'R Us [1]

2. Processing Orders

Activity diagram



Activities map nicely to the systems in WGRUS' IT department!



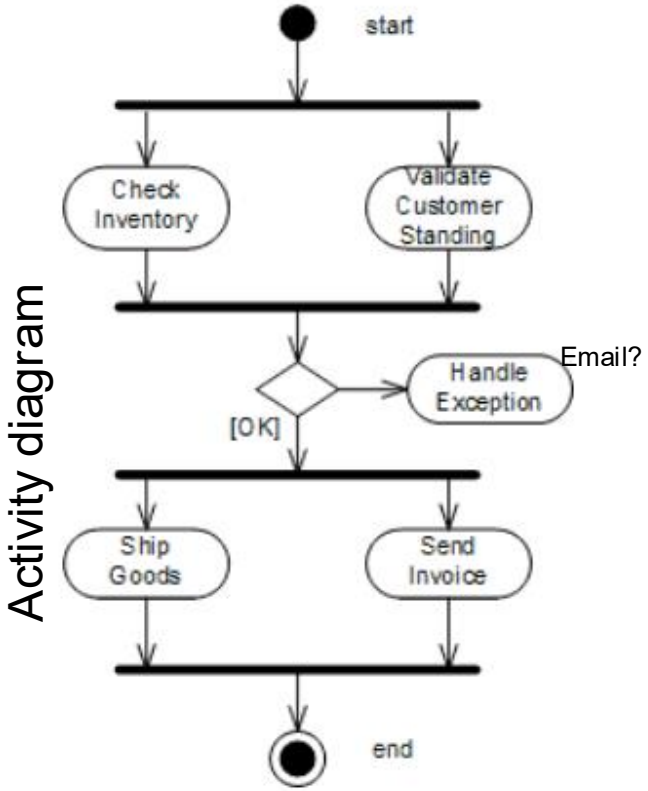
Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

2. Processing Orders

What about integration design?

- Request/Response (synch)?
- Messaging (asynch)?
- ...?



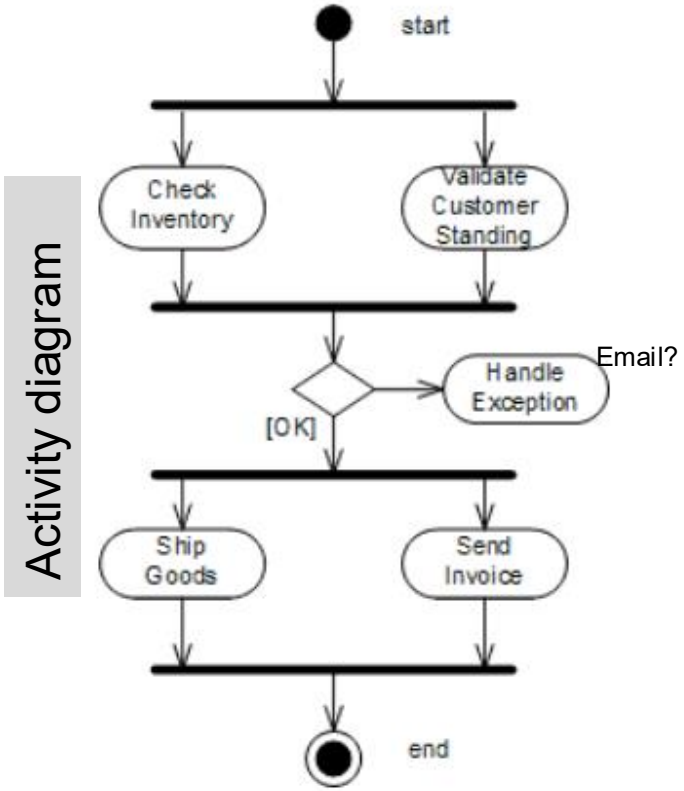
Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

2. Processing Orders

What about integration design?

- **Asynchronous/Messaging!**
  - No need to wait, no need to handle comm. errors, if target is down no problem, etc…
  - Send and forget is nice!



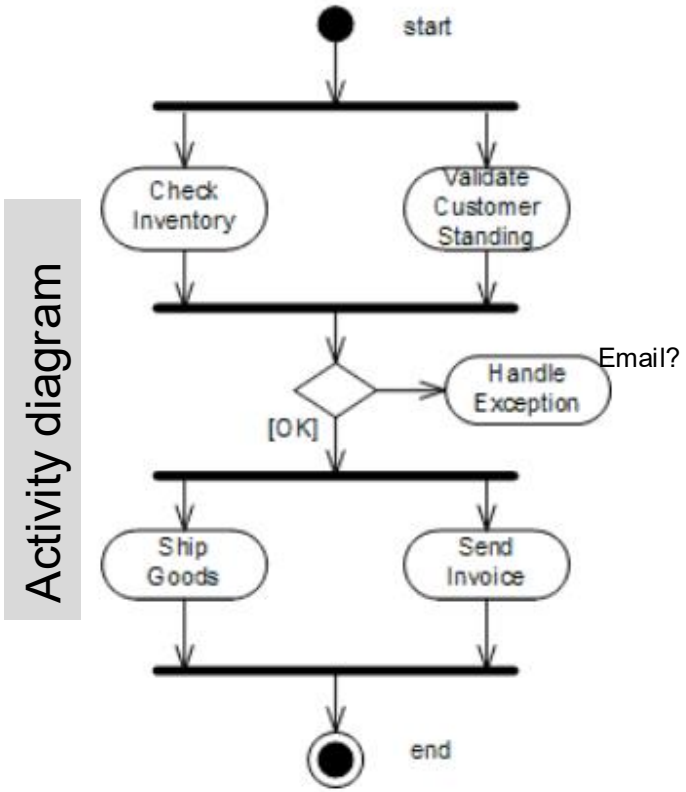
Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

2. Processing Orders

What about integration design?

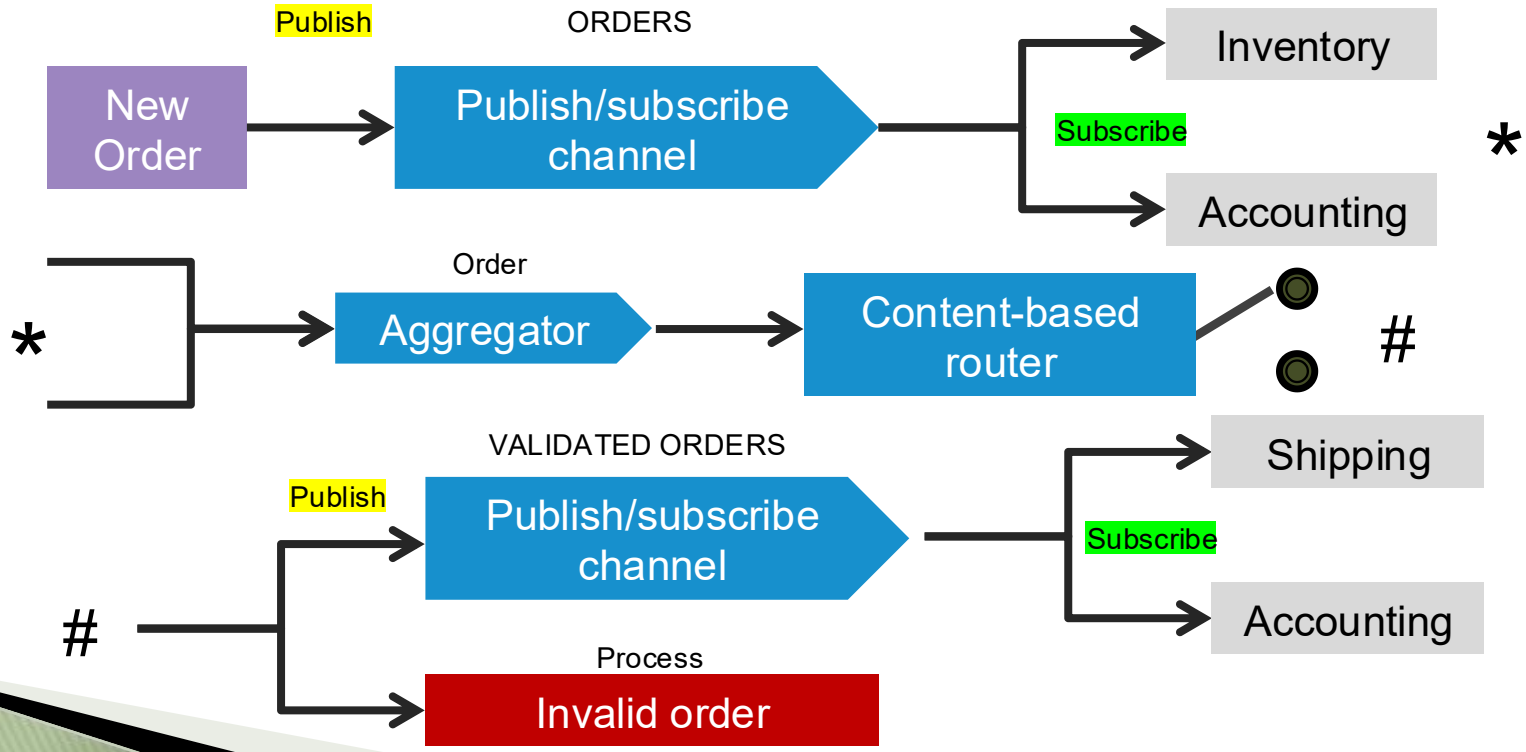
- Asynchronous/Messaging!
  - **Publish/Subscribe Model**



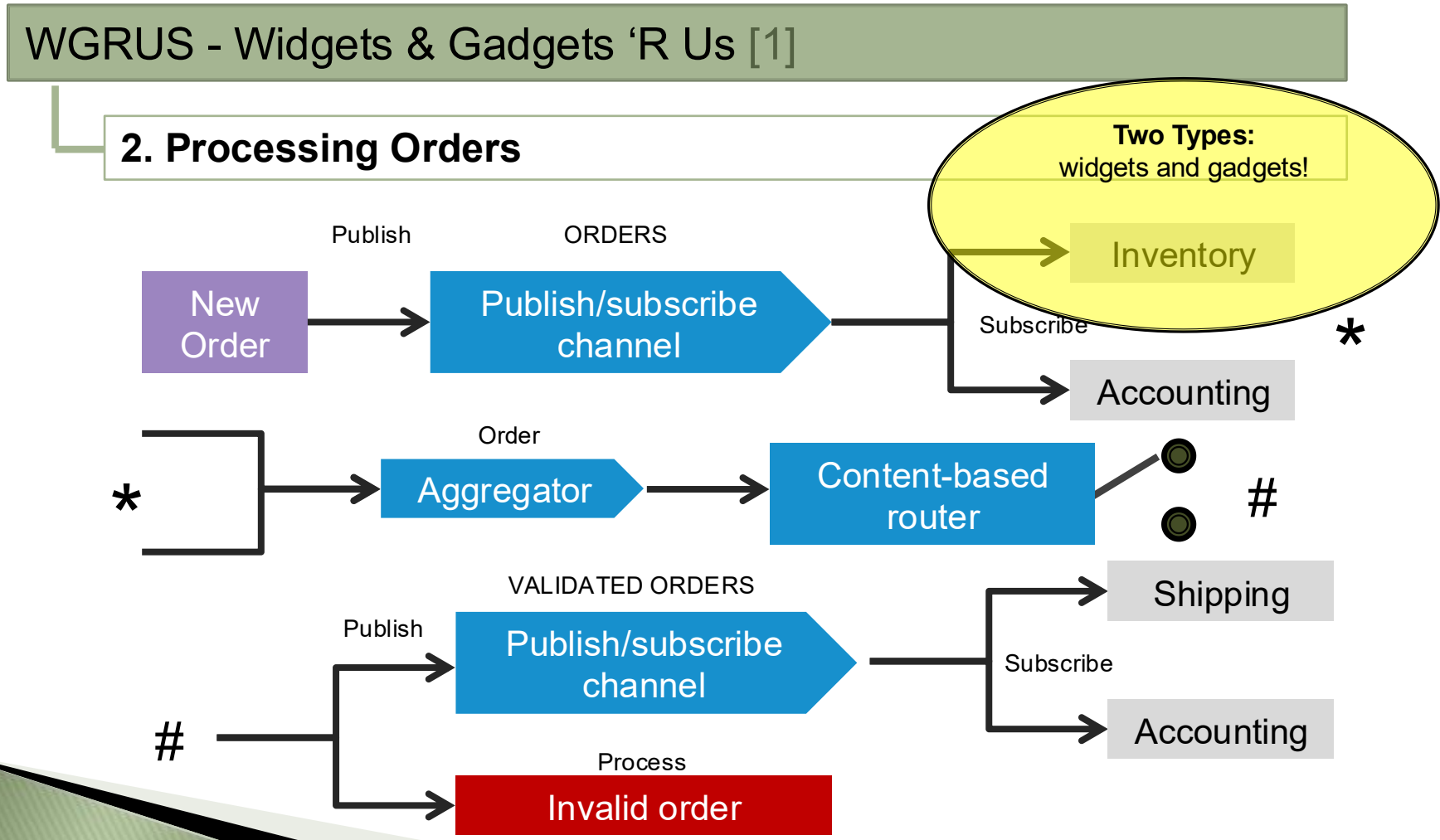
Kick-off Scenario

WGRUS - Widgets & Gadgets 'R Us [1]

2. Processing Orders



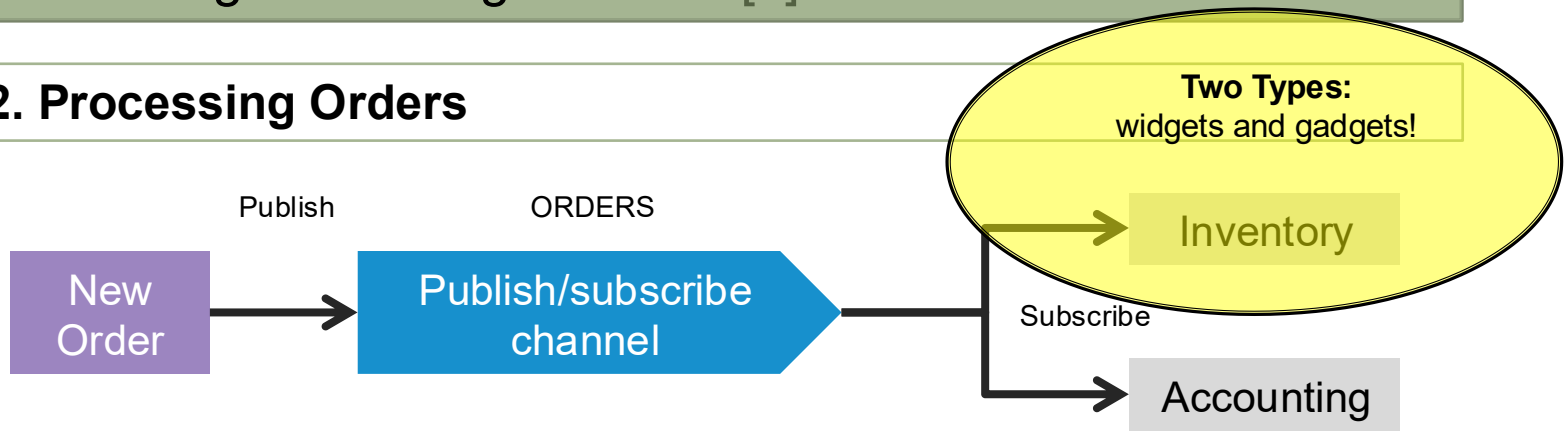
Kick-off Scenario

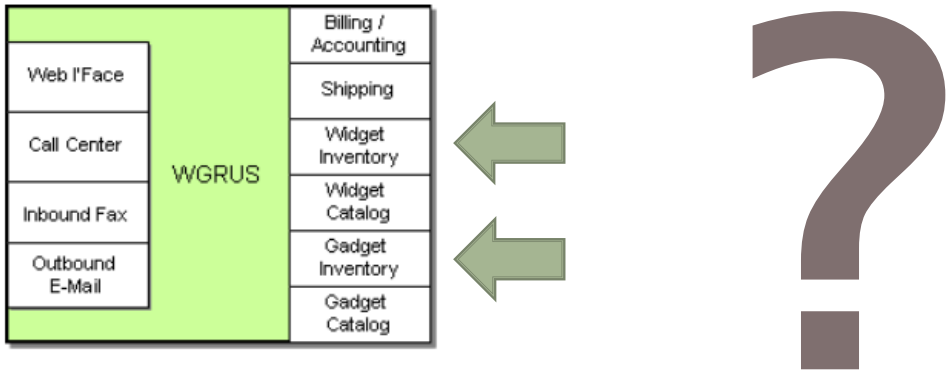


Kick-off Scenario

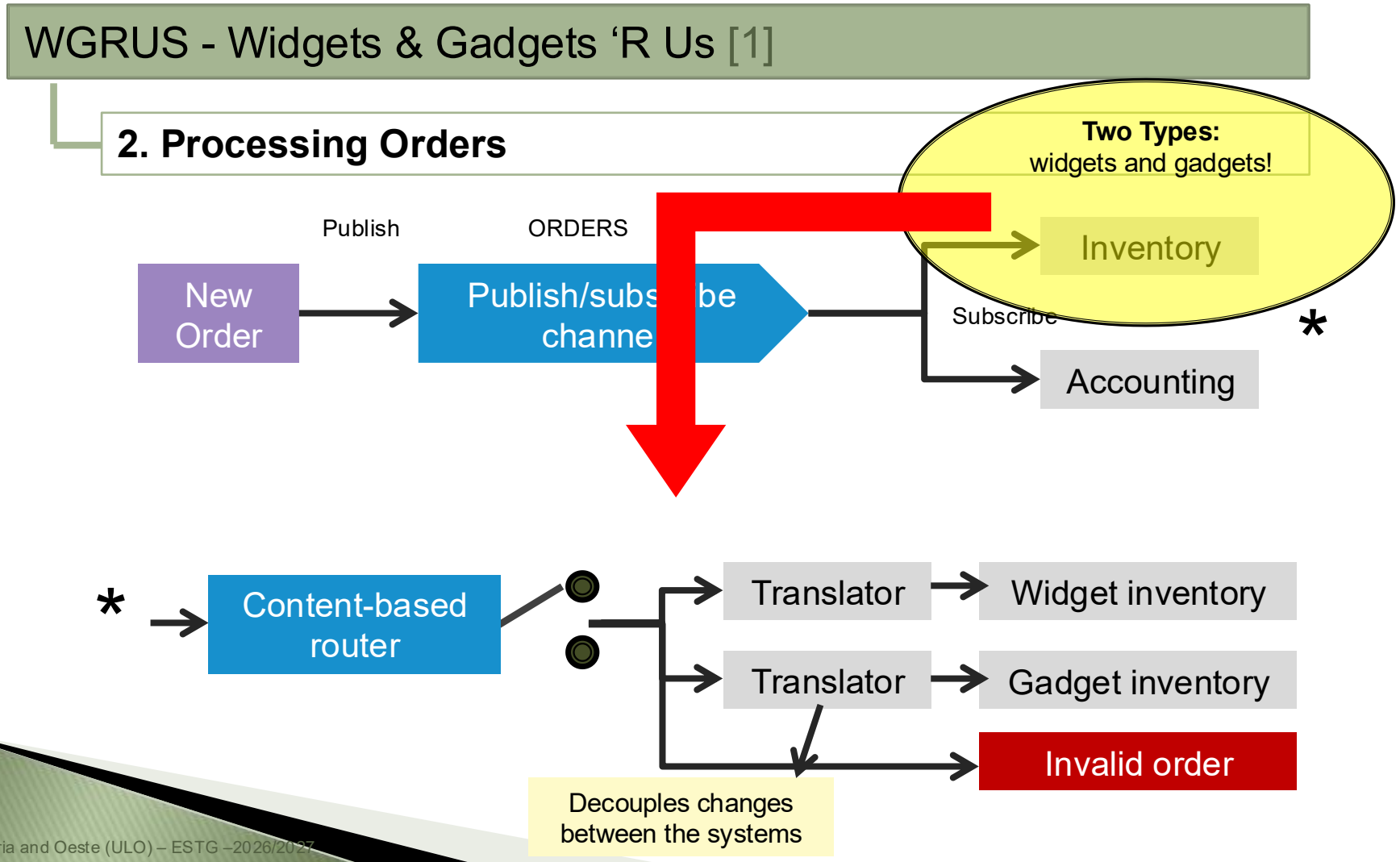
WGRUS - Widgets & Gadgets 'R Us [1]

2. Processing Orders





Kick-off Scenario



Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

2. Processing Orders

What about orders with multiple items?



Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

2. Processing Orders

What about orders with multiple items?

Splitter!

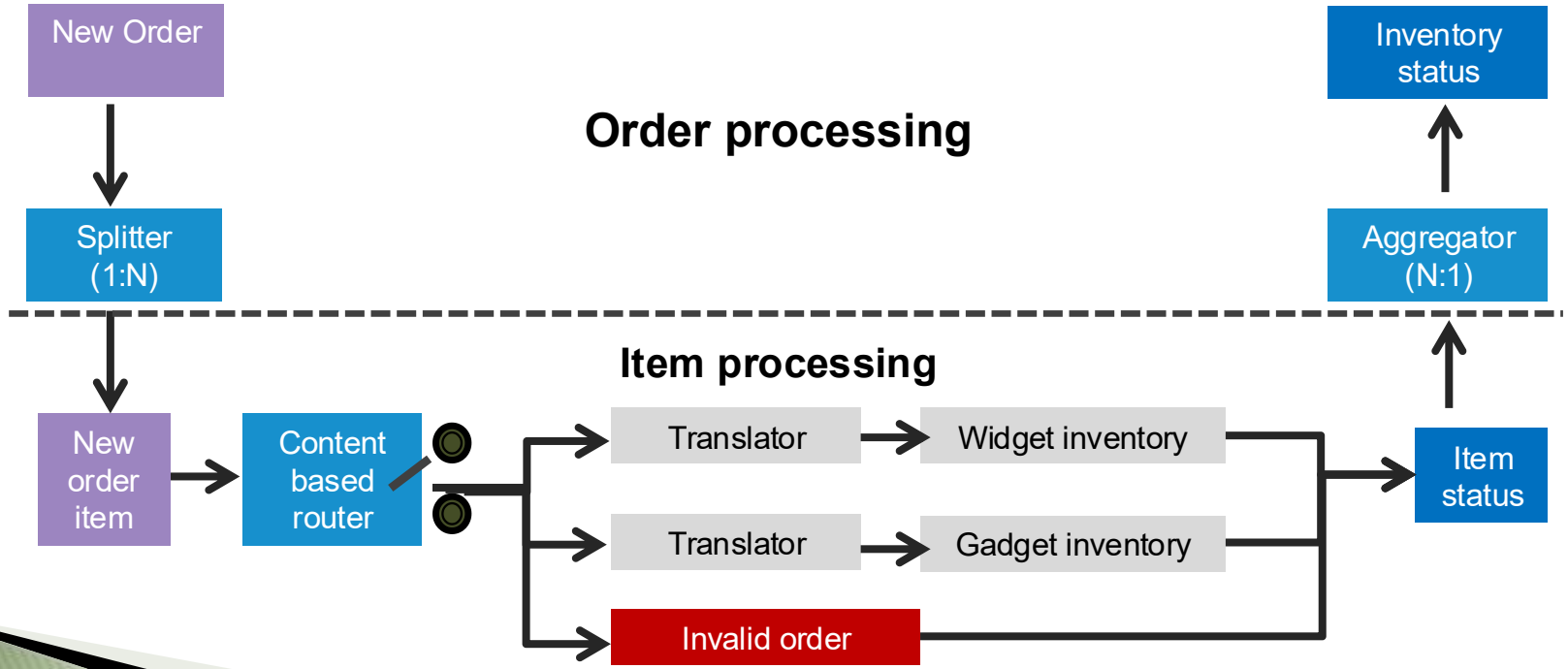
We use a splitter to split a message (order) into N messages (items)

Then, each message is routed to the right inventory system by the content-based router

Kick-off Scenario

WGRUS - Widgets & Gadgets 'R Us [1]

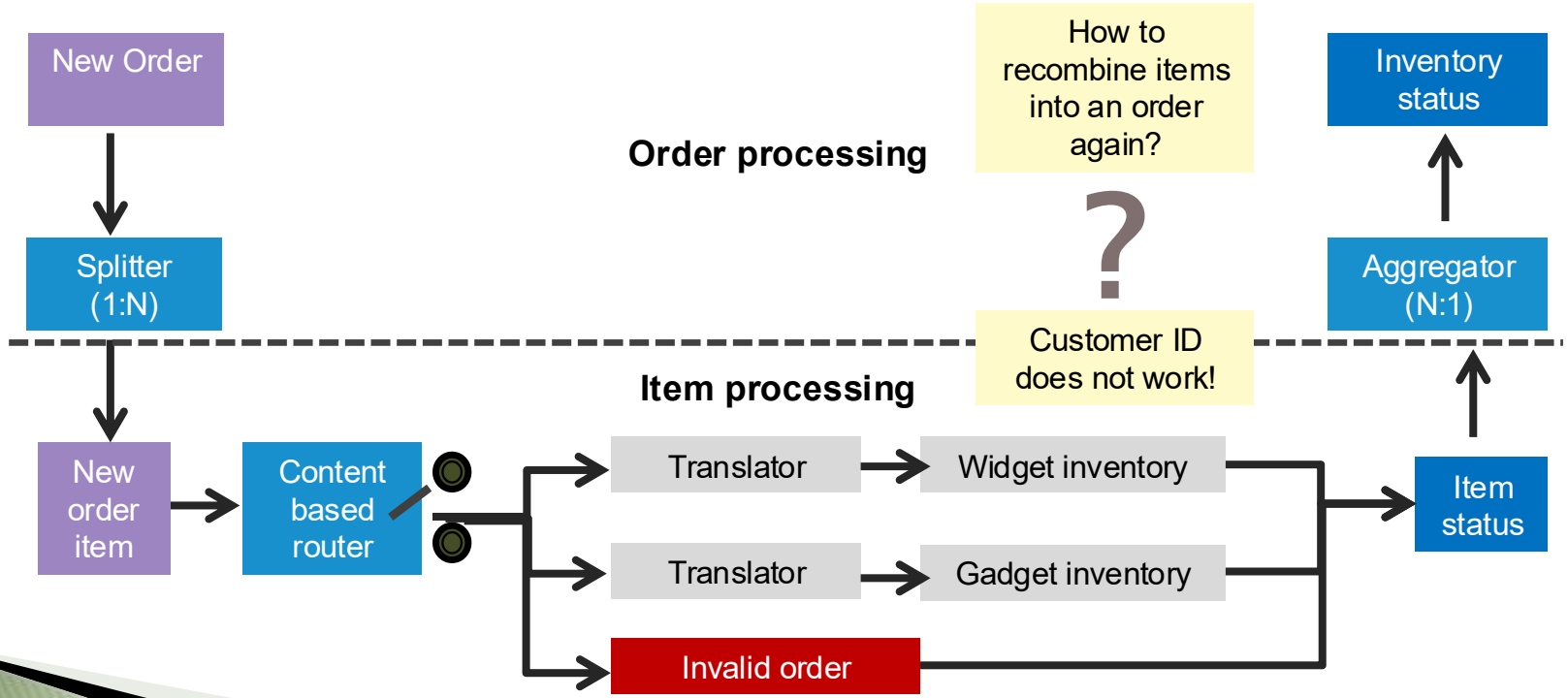
2. Processing Orders



Kick-off Scenario

WGRUS - Widgets & Gadgets 'R Us [1]

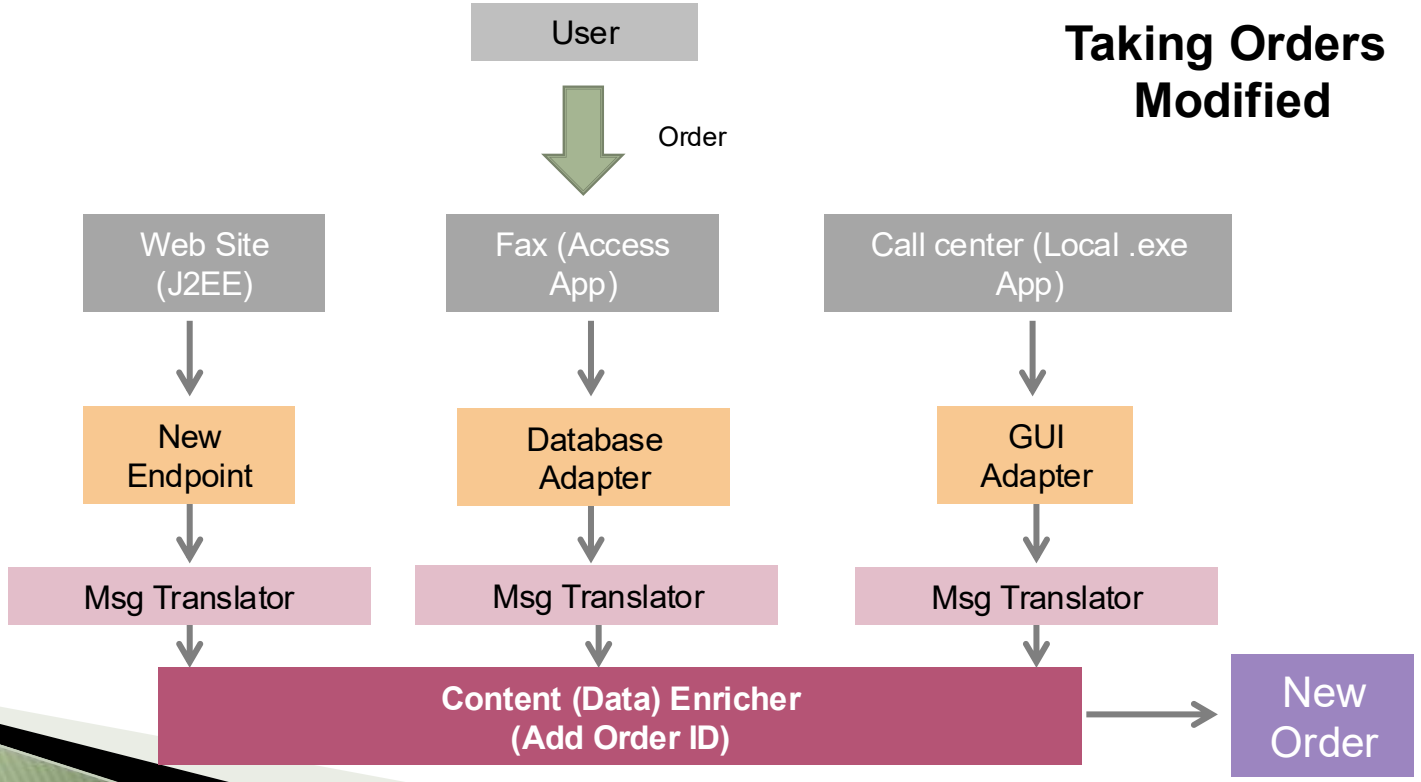
2. Processing Orders



Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

2. Processing Orders



Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

3. Checking status

Suppose now an item is out of stock…

No problem! We are using asynchronous approach! ☺

The order is naturally delayed…

Ok, but we need to check the status of a specific order…

Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

3. Checking status

We need to know the last message related to the order

As we are using the **Publish/Subscribe** Model…

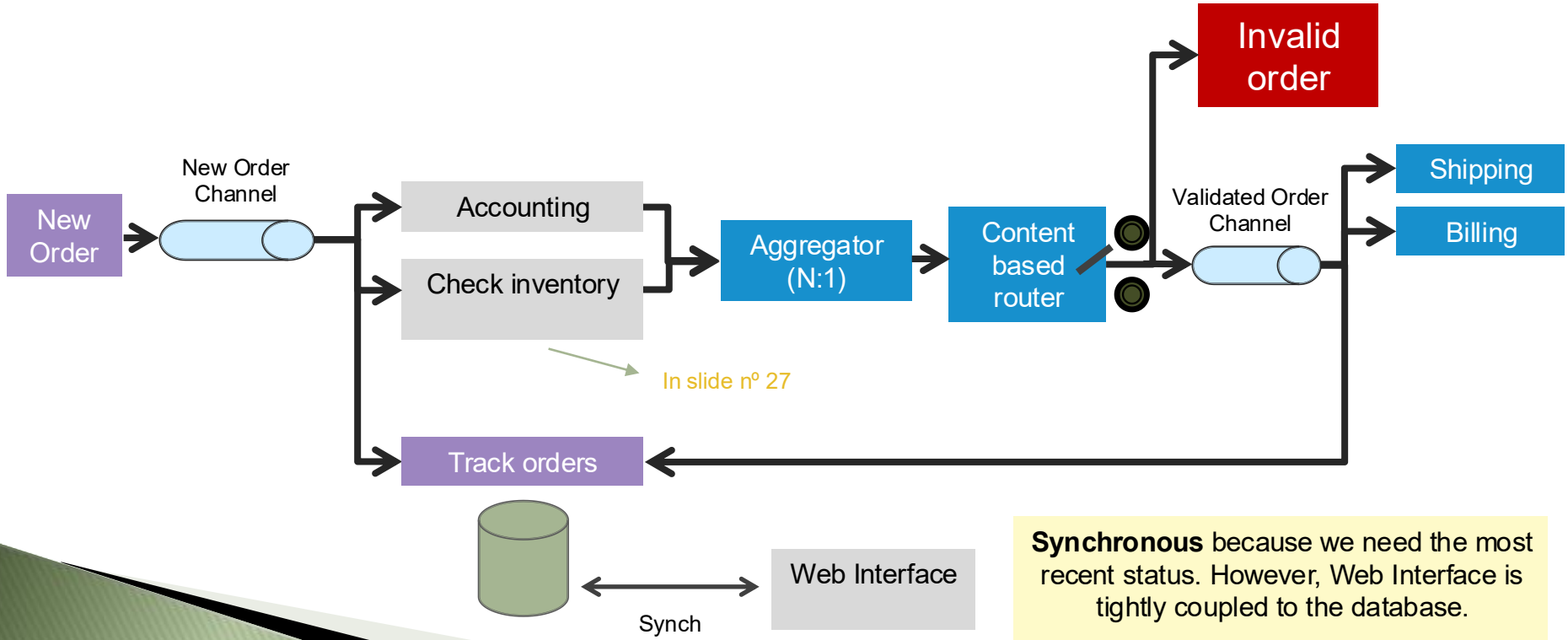
Simply add a new **subscriber** ☺

The subscriber saves messages in a database

Kick-off Scenario

WGRUS - Widgets & Gadgets 'R Us [1]

3. Checking status



Synchronous because we need the most recent status. However, Web Interface is tightly coupled to the database.

Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

4. View catalog

View widgets and gadgets in the same web site (single view)

Since catalog updates (suppliers) occur every **3 months**…

It doesn´t make sense to use a PUB/SUB

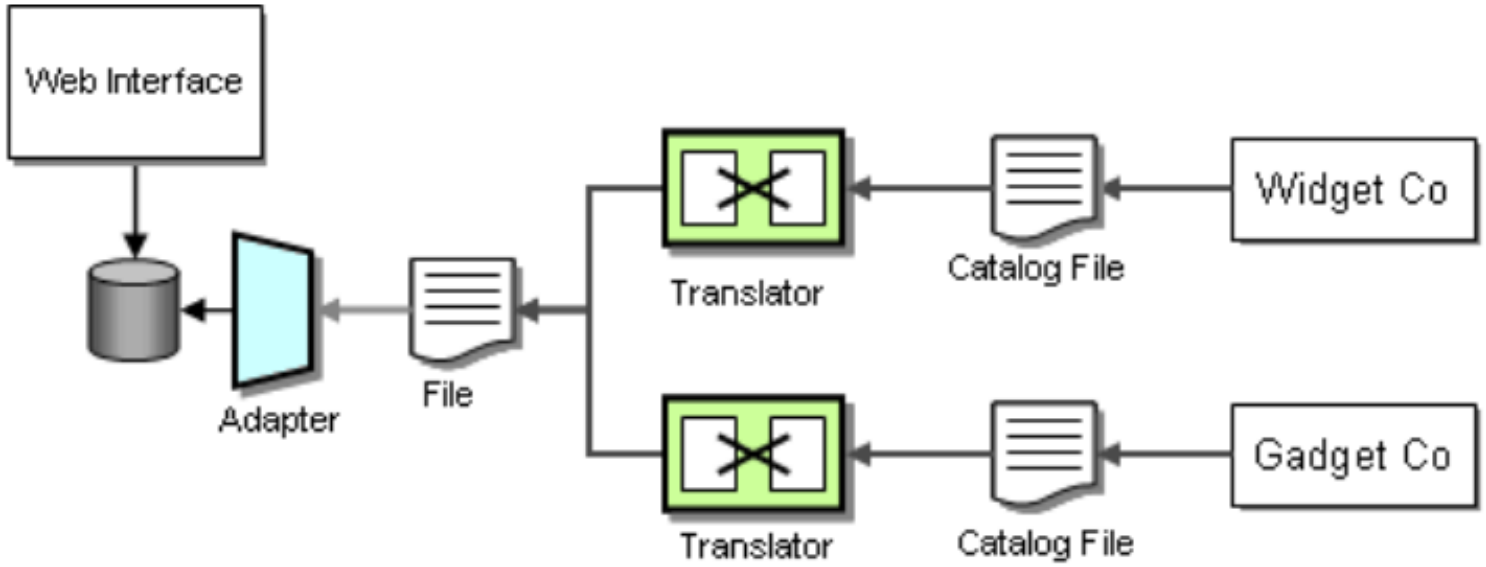
Public Internet? Firewalls, etc…

File downloading from suppliers, via FTP!

Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

4. View catalog

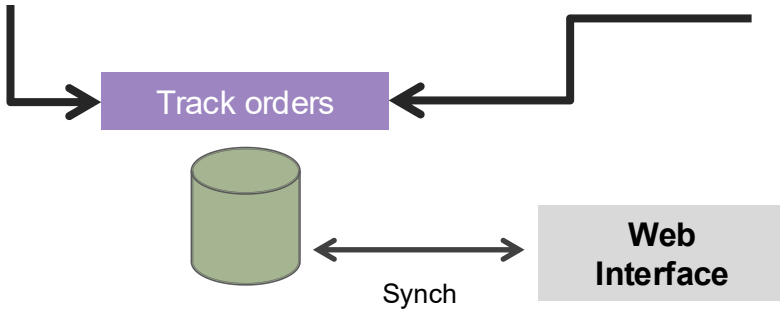


Updating catalog via file transfer

Kick-off Scenario

WGRUS - Widgets & Gadgets ‘R Us [1]

- **5. System monitoring**
  - The **tracking orders** database is a good source of information (metrics)
    - E.g., average time to fulfill an order
    - Etc.



Kick-off Scenario

- WGRUS - Widgets & Gadgets ‘R Us [1]
  - **Summary**
    - Different used integration strategies:
      - File Transfer
      - Shared Database
      - Asynchronous Messaging

References/Bibliography

[1] Gregor Hohpe, and Bobby Woolf, “Enterprise Integration Patterns: Designing Building, and Deploying Messaging Solutions”, Pearson Education, Inc., 2003